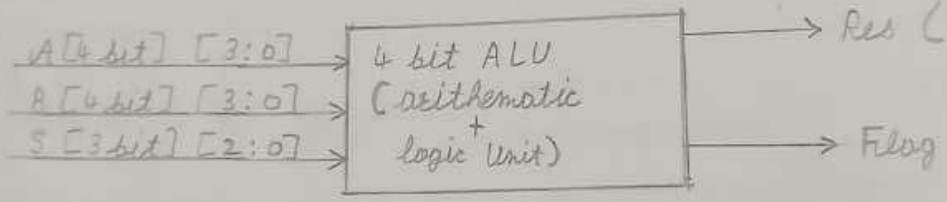
Draw the entity block diagram and explain the significance of I/O Ports of 4-bit ALU. Mention the datatype of ALU ports along with number of operations.



Port Name	Direction	Width	Data - type	Description
A	Input	4 bits	STD_Logic_VECTOR (3 down to 0)	first operand of ALU.
B	Input	4 bits	STD-Logic-VECTOR (3 down to 0)	Second operand of ALU.
S	Input	3 bits	STD-Logic-VECTOR (2 down to 0)	operation Selection line
Result	output	4 bits	STD-logic_VECTOR (3 down to 0)	Final result after operation
Flag	output	1 bits	STD-Logic	Carry / overflow flag.

cedure:

1. Open Vivado 18.2 tool.
2. Create a new VHDL project.
3. Select target FPGA device/board for impl
4. Write VHDL code in editor window of V
5. Simulate the code using test bench.
6. Synthesize the design and assign the pa
7. Run implementation and generate bit s
8. Connect the hardware and program th
9. Observe the results on the target boar

ults: Attach the following documents

1. VHDL Code Print
2. VHDL Test Bench code Print
3. Simulation Waveforms Print (On
4. Synthesis Report Print (Selected
5. UCF file Print

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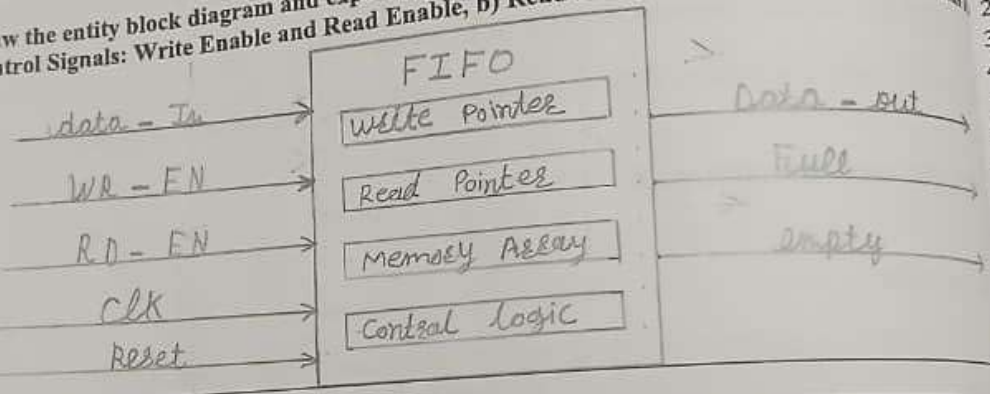
estions:

) Write the VHDL code and s

1. **Synchronous FIFO**: A synchronous FIFO has one clock domain in which read and write ... both driven by the same clock. The read and write pointers of the FIFO are updated synch... the clocks, and data is transferred between the FIFO and the external circuit synchronously w... Synchronous FIFOs are primarily used to buffer data when the rate of data transfer exce... data processing.
2. **Asynchronous FIFO**: Data read and write operations use different clock frequencies. A... FIFO helps to synchronize data flow between two systems working on different clocks.

When designing a system that requires data transfer between clock domains, determining the F... essential for efficient and reliable operation. The FIFO depth represents the size of the FIFO, wh... buffer for holding data during the transfer process. The width of a FIFO refers to the number of b... stored in each entry or slot within the FIFO. It essentially defines how much data can be written to... operation.

Draw the entity block diagram and explain the significance of following key components of FI... Control Signals: Write Enable and Read Enable, b) Read Pointer and Write Pointer, c) Status



② Significance :-
① Control signal ① Write enable :- Enable writing input into FIFO memory when the FIFO is not full.

② Read enable :- Enable reading data from FIFO memory when the FIFO is not empty.

③ Write pointer :- Indicate the memory location where next data will be written.

Read pointer :- Indicate the memory location from the data will be read. It advance after succ... read operation.

Status flag :-
① Full - indicate that FIFO ...
② Empty - indicate that FI... read.

ocedure:

1. Open Vivado 18.2 tool.
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4. Write VHDL code in editor window of Vivado...
5. Simulate the code using test bench.
6. Synthesize the design and assign the package...
7. Run implementation and generate bit stream...
8. Connect the hardware and program the targ...
9. Observe the results on the target board.

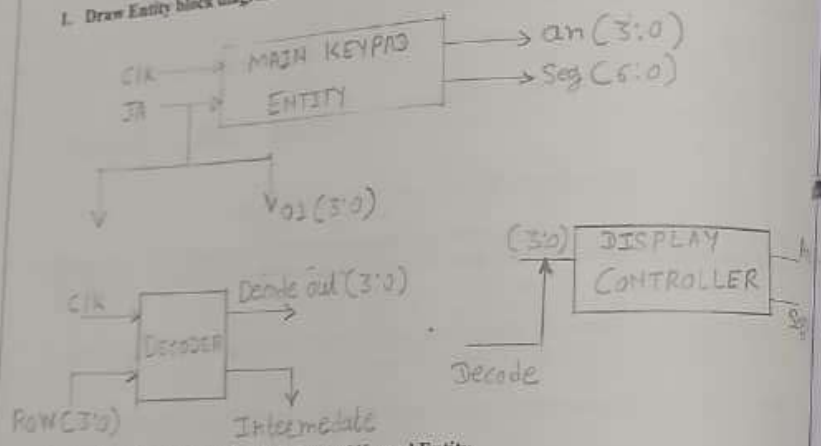
sults: Attach the following documents

1. VHDL Code Print
2. VHDL Test Bench code Print
3. Simulation Waveforms Print (On white...
4. UCF file Print

nclusion:

Thus FIFO stores Da... received. Control sig... ensure correct Dat...

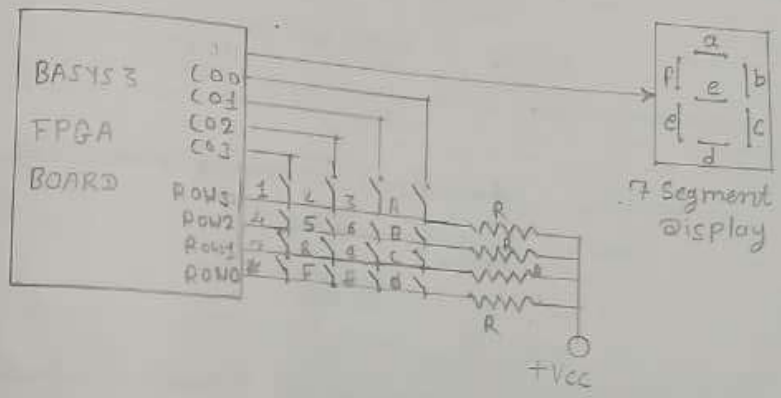
1. Draw Entity block diagram for Keypad Interface.



2. Explain the significance of I/O Ports of Keypad Entity.

I/O Port	Direction	Function
CLK	I/P	Provides 100 MHz clock Si
JA(3:0)	o/p	Selects keypad clums
JA(7:4)	I/P	Detects the pressed keyp
Decode out(3:0)	Internal	stores the decoded key Valu
An(3:0)	o/p	selects the seven-segment d
Seg(6:0)	o/p	Displays the decoded key

3. Draw Interfacing diagram of 4 X 4 matrix keypad with Artix 7 Basys 3 board.



4. Explain working of key decode logic and write calculations of delay for scanning of columns.

Counter Calculation for Keypad Scanning.

C1

$100000_{10} = 11000011010100000_2$

$F_{out} = \frac{100 \times 10^6}{100000} = 1000\,Hz.$

$T = \frac{1}{1000} = \boxed{1\,ms}$

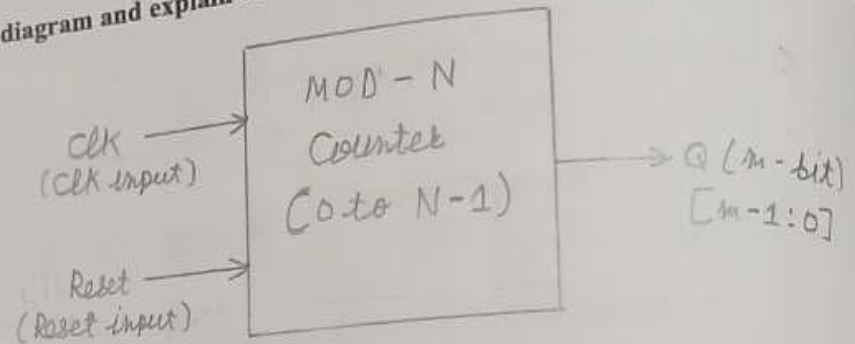
C2

$200000_{10} = 110000110101000002$

$F_{out} = \frac{100 \times 10^6}{200000} = 500\,Hz$

$T = \frac{1}{500} = \boxed{2\,ms}$

Draw the entity block diagram and explain the significance of I/O Ports of MOD-N. Mention the of Counter ports.



① Significance of I/O ports

Port	Direction	Data - type	Significance
clk	input	STD_Logic	Provide the clo
Reset	input	STD - LOGIC	Reset the counte initial state
Q	output	STD - LOGIC - VECTOR	Display the bi count.

For MOD-N

$$n = \lceil \log_{e2} N \rceil$$

② Mode of Counter:
1. Up Counter
2. Down Counter
3. up / down Counter
4. Synchronous Counter
5. Asynchronous Counter

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Record No.: PVGCOETM/ACAD/Q

rocedure:

1. Open Vivado 18.2 tool.
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2. VHDL Test Bench code Pri
3. Simulation Waveforms Prin
4. UCF file Print

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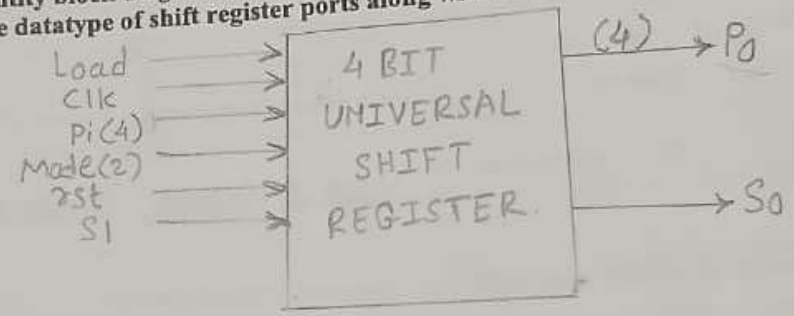
Write the VHDL code for
eform.
Write the VHDL code for 4
eform. State the difference
simulation waveform.

erences:

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Record No.: PVGCOETM/ACAD/QMS/R/ | Revision: 01 | Date:01/07/2026

Draw the entity block diagram and explain the significance of I/O Ports of 4-bit Universal shift regist
Mention the datatype of shift register ports along with modes of shift register.



1] Serial input :- Used to feed data bit-by-bit during shift left or right operation

2] Serial output :- Provides the shifted out bit after each shift operation

3] Parallel data inputs :- allows simultaneous loading of 4 bit data i register.

4] Parallel data out :- provides stored content of the register afte operation.

5] Mode Control Inputs :- Select the operation mode (hold, left, right, shift, parallel load).

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ecord No.: PVGCOETM/ACAD/

plain the process to scaling the

The i/p clock fre
(50 MHz) which is

To scale it down
a binary counter

When counter re
producing a clo

For eg :- Divid
generating a
trigger the

Cou

ocedure:

1. Open Vivado
2. Create a new
3. Select target
4. Write VHD
5. Simulate th